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Sakurai

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(54) ***VACCINIUM CORYMBOSUM* L. PLANT
NAMED ‘RYOKU NH-11’**

(50) Latin Name: *Vaccinium corymbosum* L.
Varietal Denomination: **RYOKU NH-11**

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(57) **ABSTRACT**

A new and distinct variety of *Vaccinium corymbosum* L.
plant named ‘RYOKU NH-11’, characterized by having
stronger plant vigor, a more upright plant growth habit,
comparatively early fruit ripening time, comparatively large
and uniform fruit size, and smaller and dry stem scar, as
compared to other *Vaccinium corymbosum* L. varieties.

8 Drawing Sheets

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The latin name of the genus and species of the novel
variety disclosed herein is: *Vaccinium corymbosum* L.

The novel variety of the *Vaccinium corymbosum* L. dis-
closed herein has been given the variety denomination:
‘RYOKU NH-11’.

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to Japanese Plant Breed-
ers’ Rights Application No. 31723, filed Jan. 4, 2017, the
contents of which are incorporated herein by reference in
their entirety.

BACKGROUND OF THE INVENTION

The present invention relates to a new and distinct hybrid
variety of northern highbush blueberry (*Vaccinium corym-
bosum* L.) named ‘RYOKU NH-11’. This novel variety was
found by open pollination of ‘Chandler’, a seed parent
variety, in the tests conducted for the period from 2003 to
2007 in Matsumoto-City, Nagano-prefecture, Japan. As
stated below, ‘RYOKU NH-11’ has apparently different
characteristics from those of the varieties ‘Chandler’ and
‘Blueray’, both being widely planted and being important
varieties in the Chubu district of Japan.

SUMMARY OF THE INVENTION

Blueberry variety ‘RYOKU NH-11’ exhibits outstanding
and distinguishing characteristics when grown under normal
horticultural conditions in the area from Nagano-prefecture
to the north of the Kanto in Japan, including:

- (1) stronger plant vigor and more upright plant growth
habit;

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- (2) comparatively early fruit ripening time (on average,
around July 15 of each year, in Matsumoto-city,
Nagano-prefecture, Japan);
- (3) comparatively large and uniform fruit size; and
- (4) smaller and dry stem scar.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying colored photographs (FIGS. 1 to 8)
show typical bush, flower, fruit and leaf characteristics for
the new *Vaccinium corymbosum* L. plant ‘RYOKU NH-11’.
Colors shown are as accurate as can be reasonably repro-
duced by photographic means. In some cases, the color
might differ slightly from the colors of ‘RYOKU NH-11’
recited in the description.

FIG. 1 shows a tree body of ‘RYOKU NH-11’ (photo-
graphed date: Jul. 10, 2015; photographed location: Matsu-
moto-City, Nagano-prefecture, Japan).

FIG. 2 shows a panoramic view of the cultivation area of
‘RYOKU NH-11’ (photographed date: Sep. 7, 2012; photo-
graphed location: same as FIG. 1).

FIG. 3 shows whole flowers of ‘RYOKU NH-11’ (pho-
tographed date: May 6, 2014; photographed location: same
as FIG. 1).

FIG. 4 shows broken-down parts of a flower of ‘RYOKU
NH-11’ (photographed date: May 6, 2014; photographed
location: same as FIG. 1).

FIG. 5 shows fruits of ‘RYOKU NH-11’ (photographed
date: Jul. 20, 2012; photographed location: same as FIG. 1).

FIG. 6 shows a cross-section of the fruits of ‘RYOKU
NH-11’ (photographed date: Jul. 20, 2012; photographed
location: same as FIG. 1).

FIG. 7 shows an upper side of the leaves (10 sheets) of 'RYOKU NH-11' (photographed date: Aug. 3, 2012; photographed location: same as FIG. 1).

FIG. 8 shows a lower side of the leaves (10 sheets) of 'RYOKU NH-11' (photographed date: Aug. 3, 2012; photographed location: same as FIG. 1).

DETAILED BOTANICAL DESCRIPTION

A. Distinctive Characteristics of 'RYOKU NY-11'

As described above, 'RYOKU NH-11' was obtained by the open pollination of 'Chandler', a seed parent (Note: the possible pollen parent might be 'Blueray'). On Apr. 15, 2003, about 10,000 seeds of 'Chandler', which were cultivated in a field in Matsumoto-City, Nagano-prefecture, Japan, were seeded in plant seeding trays, and then transplanted to pots. The resulting seedlings (about 3,000) were planted in cultivation pots on May 1, 2004. Fructification of the planted seedlings were started from Jun. 20, 2007 (on Year 3), and about 200 plant individuals were selected based on the characteristics, including larger fruit size, better taste balance of sweetness and acidity, higher yield per plant, earlier ripening time, large and uniform fruits, etc. For the about 200 plant individuals selected, test plots (each including about 20 cuttings per plant individual) were formed, where these plants were asexually propagated by cutting means. During the period from Jul. 20, 2009 to Dec. 25, 2016 and for 3 generations, the plants were propagated and examined for their characteristics based on the growth, yield, and fruits quality in each test plot. For 10 test plots, the characteristics of the plants were observed for the period between the flowering time and the ripening time yearly for 5 years, and since neither variant nor off-type plant was observed for the period, the characterization of 'RYOKU NH-11' was finished on Dec. 25, 2016 and the breeding was completed.

'RYOKU NH-11' is a blueberry clone distinguishable from the important blueberry varieties 'Chandler' and 'Blueray', both of which are widely planted in the Chubu district of Japan, due to its characteristics including stronger plant vigor, more upright plant growth habit, earlier fruit ripening time, larger and more uniform fruit size, and smaller and dry stem scar. 27 plants of 'RYOKU NH-11' had been propagated by cutting means in Matsumoto, Nagano, Japan, and all the resulting plants were phenotypically indistinguishable from the original plant variety 'RYOKU NH-11'. In addition, comparing to 'Chandler' and 'Blueray', the claimed plant 'RYOKU NH-11' has a more upright plant growth habit, about 1-week earlier fruit ripening time (on average, around July 15 of each year in Matsumoto, Nagano, Japan), more uniform fruit size and smaller stem scar when compared to its related variety 'Chandler', and has stronger plant vigor, larger fruit size, and smaller stem scar when compared to the 'Blueray' variety (see Table 1 below).

The following data defining the characteristics of 'RYOKU NH-11' were collected from the asexual propagation carried out in Matsumoto, Nagano, Japan. The plant history was taken on a plot of 10 five-year-old plants growing in Matsumoto, Nagano, Japan. 'RYOKU NH-11' has not been observed under all possible environmental conditions, and the measurements provided might therefore vary if grown in different environments. Where averages are given, the sample size was 10.

B. Phenotypic Description of *Vaccinium corymbosum* L. ('RYOKU NH-11')

Characteristics of 'RYOKU NH-11' are further specifically described as follows:

1. Plant:

Plant vigor.—Strong, where the plant vigor is stronger than 'Blueray'.

Plant size.—Large.

Growth habit.—Upright.

Plant height.—1.7 m on average for 5-year old plant.

Plant spread.—1.1 m on average for 5-year old plant.

Color of bark of plant.—Deep Red, 185-A (The R.H.S. Colour Chart).

Tendency toward evergreenness.—Absent.

Cold hardiness.—Survived in winter frost (below -10° C.) with minimum damage.

Ease of propagation.—At least 70% rooting percentage in each of the dormant wood cutting and softwood stem cutting, comparable to the other varieties.

2. Trunk and branches:

Suckering tendency.—Less suckering as in 'Chandler'.

Surface texture (of 6-month-old shoots).—Medium smoothness.

Surface texture (of 3-year-old and older wood).—Medium smoothness.

Color of new twigs observed in the field.—Yellow green.

Internode length.—15.3 mm on average.

3. Leaves:

Length including petiole.—69.4 mm on average.

Width at widest point.—35.2 mm on average.

Shape.—Elliptic.

Leaf margin.—Entire.

Color.—Upper surface of leaves: Moderate Green, 135-B. Lower surface of leaves: Moderate Olive Green, 137-B (The R.H.S. Colour Chart).

Pubescence.—Upper Surface of leaves: Absent. Lower Surface of leaves: Absent. Margins: Absent.

Timing of vegetative bud burst.—Medium.

4. Flowers:

Shape.—Campanulate.

Color of opened flower.—Greenish White, 157-D (The R.H.S. Colour Chart).

Flowering period.—Mean date of 50% opening of flowers in Matsumoto-City, Nagano-prefecture, Japan is May 1 (2 days earlier than 'Chandler').

Corolla.—Diameter: 9.6 mm on average. Length (from pedicel attachment point to corolla tip excluding the pedicel): 12.3 mm on average. Color: light yellow white. Anthocyanin coloration in corolla tube — Absent or very weak.

5. Reproductive organs:

Pollen.—Color: Yellow.

6. Fruit:

Mean date of 50% harvest in Matsumoto-City, Nagano-prefecture.—July 15.

Diameter of calyx aperture on mature berry.—7.2 mm on average.

Size and shape of calyx lobe on mature berry.—Medium in size, outcurving, and having deep calyx basin.

Detachment force for ripe berries (easy, medium, hard).—Easy.

Fruit cluster density (sparse, medium, dense).—Medium.

Fruiting type.—On one-year old shoots only.

7. Berry:

Cluster (tight, medium, loose).—Medium.
Weight (on well-pruned plants).—4.51 g on average.
Height.—15.4 mm on average.
Width.—22.0 mm on average.
Shape.—Oblate.
Skin of fruit, with bloom.—Light Purplish Blue, 98-D (The R.H.S. Colour Chart).
Intensity of fruit bloom.—Medium.
Skin of fruit, without bloom.—Greyish Purplish Blue, 103-A (The R.H.S. Colour Chart).
Immature berry color, with bloom.—Light green.
Immature berry color, without bloom.—Light yellow green.
Flesh color.—Pale Yellow Green, 149-C (The R.H.S. Colour Chart).
Peel color.—Medium blue.
Color of seeds.—Brownish Orange, N167-B (The R.H.S. Colour Chart).
Pedicel scar.—Medium, 2.54 mm on average.
Firmness.—Medium.
Intensity of fruit sweetness.—Medium, Bx 10.8.
Intensity of fruit acidity.—High, pH 2.80.
Texture.—Crispy and juicy, medium seeds.

8. Use: ‘RYOKU NH-11’ produce northern highbush blueberries suitable for fruit-picking farms, fresh fruit markets and processed fruit markets, etc.

9. Resistance to disease, insects, and mites: ‘RYOKU NH-11’ grew vigorously and showed excellent bush survival in the field. It appears to be tolerant to stem blight (*Botryosphaeria* spp.) and root rot (*Phytophthora cinnamoni*), with very few young plants dying soon after planting. The response of ‘RYOKU NH-11’ to the various fungal species that cause summer leaf spots is typical of other northern highbush varieties, and fungicide applications may be needed after harvest in order to reduce foliar diseases and to retain leaves until autumn and make maximum flower bud set. Similarly, susceptibility to typical blueberry insect and mite pathogens, such as spotted wing *Drosophila* (*Drosophila suzukii*), blueberry gall midge (*Dasineura oxycoccana*) and blueberry bud mite (*Acalitus vaccini*), is similar to other northern highbush cultivars.

TABLE 1

(Comparison of characteristics among varieties)				
Charact. No	UPOV No.	Code	Characteristics	Definition
1	1 (*)	QN (+)	Plant: vigor	Strength of growth level of plant
2	2 (*)	QN	Plant: size	Size of plant crown
3		QN G	Plant: growth habit	Whole shape of plant without pruning during dormant period
4	3	PQ	One-year-old shoot: color	Color of middle part of shoot extended before dormant period
5	4	QN	One-year-old shoot: length	Length of middle part of shoot extended before dormant period
6		QN	One-year-old shoot: length of internode (upper half)	Length of internode of shoot extended before dormant

TABLE 1-continued

(Comparison of characteristics among varieties)				
7	5 (*)	QN	Leaf: length	period (upper half) Length of leaf sufficiently expanded
8	6	QN	Leaf: width	Maximum width of mature leaf
9	7	QN	Leaf: ratio length/width	Ratio of leaf length to maximum width (leaf length/leaf width)
10	8 (*)	PQ	Leaf: shape	Shape of mature leaf
11	9	QN (+)	Leaf: shape of tip	Shape of lip of mature leaf
12		QL	Leaf: color of upper side	Color of surface of mature leaf
13	10 (*)	QN	Only varieties with green leaf color: Leaf, intensity of green color on upper side	Intensity of green color on surface of mature leaf
14	11 (*)	QL	Leaf: margin	Type of margin of mature leaf
15	12	QN	Flower bud: anthocyanin coloration	Intensity of anthocyanin coloration of flower bud occurring to one year old shoot
16	13	QN	Inflorescence: length (excluding peduncle)	Length of inflorescence at flowering time (excluding peduncle)
17	14	PQ	Flower: shape of corolla	Shape of corolla at full bloom
18	15 (*)	PQ	Flower: color of corolla	Color of corolla at full bloom
19		QN	Flower: size of corolla tube	Size of corolla tube at full bloom
20	16 (*)	QN	Flower: anthocyanin coloration of corolla tube	Intensity of anthocyanin coloration on surface of corolla tube
21	17	QL	Flower: ridges on corolla tube	Presence or absence of ridges on corolla tube
22	18	QN	Fruit cluster: density	Density of fruit per fruit cluster
23	19 (*)	QN	Unripe fruit: intensity of green color	Intensity of green color of fruit before ripening
24	20 (*)	QN	Fruit: size	Size of fruit at ripening
25	21 (*)	PQ (+)	Fruit: shape in longitudinal section	Shape in longitudinal section of fruit at ripening
26	22	QN (+)	Fruit: size of scar	Size of stem scar of mature fruit
27		PQ (+)	Fruit: shape of calyx cavity	Shape of calyx cavity of mature fruit
28	23	QN	Fruit: attitude of sepals	Attitude of sepals relative to mature fruit
29	24	QN	Fruit: type of sepals	Direction of curving of sepals
30	25	QN	Fruit: diameter of calyx basin	Diameter of calyx basin of mature fruit
31	26	QN	Fruit: depth of calyx basin	Depth of calyx basin of mature fruit

TABLE 1-continued

(Comparison of characteristics among varieties)					
Charact. No	Method	Class	State	Standard Variety (Ex tor.)	
1	Observation (a) VG	3	weak	Bluetta, Meader	45
		5	medium	Collins, Weymouth	
		7	strong	Berkeley, Homebell, Woodard	
2	Observation (a) VG	3	small	Avonblue, Bluetta, Flordablue	50
		5	medium	Bluecrop, Earliblue	
		7	large	Dixi, Homebell, Tifblue	
3	Observation (a) VG	1	upright	Becyblue, Bluechip, June, Spartan	60
		2	semi-upright	Bluecrop, Lateblue	
		3	spreading	Northland, Weymouth	
4	Observation (a) VG	1	green	Briteblue, Homebell	65
		2	greenish red		
		3	greyish red		

TABLE 1-continued

(Comparison of characteristics among varieties)					
		4	reddish yellow	Berkeley, Dixi Blueray, Darrow, Weymouth	
		5	reddish brown		
		6	dark red		
	5	Measurement	3 short	Avonblue, Weymouth	
		mm	5 medium		
		(a)	7 long		
	6	VG		Jersey	
		Observation	3 short		
		(a)	5 medium		
	7	VG	7 long		
		Measurement	3 short		
		mm	5 medium		
		(b)	7 long		
		MS/ VG			
		Measurement	3 narrow		
	8	mm	5 medium		
		(b)	7 broad		
		MS/ VG			
	9	Measurement	3 small		
		(b)	5 medium		
		MS/ VG	7 large		
	10	Observation	1 lanceolate	Northland Berkeley, Collins, Coville	
		(b)	2 ovate		
		VG	3 elliptic		
	11	Observation (b) VG	4 oblong	Weymouth, Woodard Earliblue, Tifblue Berkeley, Climax, Southland	
			3 acute		
			5 medium		
	12	Observation (b) VG	7 obtuse		
			1 yellow	Bluechip, Bluecrop, Blueray	
			2 green		
	13	Observation (b) VG	3 light		
			5 medium		
			7 dark		
	14	Observation (b) VG	1 entire		
			2 serrate		
	15	Observation (a) VG	3 weak	Bluecrop, Jersey Northblue, Northsky	
			5 medium		
			7 strong		
	16	Measurement mm (c) MS/ VG	3 short	Aliceblue, Bluetta, Briteblue Avonblue, Berkeley, Bluecrop Blueray, Collins, Coville Bluebell, Delite, Dixi, Tifblue	
			5 medium		
			7 long		
	17	Observation (c) VG	1 urceolate		
			2 campanulate		
			3 cylindrical		
	18	Observation (c) VG	1 white		
			2 creamy white		
			3 greenish white		
	19	Observation (c) VG	4 light pink		
			3 small		
			5 medium		
	20	Observation (c)	7 large		
			1 absent or very weak		
			3 weak		

TABLE 1-continued

(Comparison of characteristics among varieties)				
	VG	5	medium	
		7	strong	
21	Observaton	1	absent	Herbert
	(c)	9	present	Aliceblue
	VG			Berkeley, Dixi
22	Observation	3	sparse	Homebell,
	(d)			Jersey,
	VG	5	medium	Woodard
				Bluechip,
				Bluecrop,
				Bluetta
		7	dense	Darrow,
				Herbert, Patriot
23	Observation	3	light	
	VG	5	medium	
		7	dark	
24	Observation	3	small	Homebell,
	(d)			June,
	VG			Northblue
		5	medium	Collins,
				Earliblue
		7	large	Berkeley,
				Bluecrop, Spartan
25	Observation	1	elliptic	
	(d)	2	round	Berkeley,
	VG			Jersey,
				Sharpblue
		3	oblate	Earliblue,
				Harison,
				Woodard
26	Observation	3	small	
	(d)	5	medium	
	VG	7	large	
27	Observation	1	star	
	(d)	2	circular	
	VG			
28	Observation	1	erect	
	(d)	2	erect to semi-erect	
	VG	3	semi-erect	
		4	level	
29	Observation	1	incurving	
	(d)	2	straight	
	VG	3	reflexed	
30	Observation	3	small	Avonblue,
	(d)			Bluechip,
	VG			Sharpblue
		5	medium	Blueray,
				Woodard
		7	large	Coville, Darrow,
				Homebell
31	Observation	3	shallow	Bluecrop,
	(d)			Rancocas,
	VG			Tifblue
		5	medium	Earliblue, Jersey
		7	deep	Blueray,
				Collins, Dixi
32	Observation	1	very weak	
	(d)	3	weak	Dixi, Herbert,
	VG			Sharpblue
		5	medium	Collins, Coville
		7	strong	Avonblue,
				Bluecrop,
				Tifblue
33	Observation	1	light blue	Berkeley,
	(d)			Bluechip,
	VG			Tifblue
		2	medium blue	Blueray,
				Jerse, June
		3	dark blue	Dixi, Homebell
		4	blue red	
34	Observation	3	soft	Herbert,
	(d)			Homebell,
	VG/			Spartan

TABLE 1-continued

(Comparison of characteristics among varieties)				
	VS	5	medium	Collins, Dixi
		7	firm	Coville,
				Southland
		9	very firm	
35	Observation	1	white	Berkeley,
	(d)			Bluecrop,
	VG			Blueray
		2	cream	Earliblue
		3	light green	Bluechip,
				Lateblue,
				Sharpblue
		4	light purple	Aliceblue,
				Delite,
				Homebell
36	Observation	3	low	Avonblue,
	(d)			Bluechip
	VG	5	medium	Berkeley,
				Bluetta, Spartan
		7	high	Aliceblue,
				Bluecrop,
				Blueray
37	Observation	3	low	Earliblue,
	(d)			Homebell
	VG	5	medium	Blueray, Herbert
		7	high	Collins,
				Elliott, Lateblue
38	Observation	1	on one-year-old	
	(c)		shoots only	
	VG	2	on one-year-old	
			and current	
			season's shoots	
			less	
39	Observation	3		Earliblue,
	(d)			Herbert,
	VG			Spartan
		5	medium	Avonblue,
				Berkeley,
				Bluechip
		7	much	Briteblue, Climax,
				Darrow
40	Measurement	3	early	Avonblue,
	MG			Beckyblue,
				Sharpblue
		5	medium	Darrow,
				Weymouth
		7	late	Elliott, Lateblue
41	Measurement	1	very early	
	MG	3	early	
		5	medium	Bluecrop,
				Collins, Woodard
		7	late	Dixi, Herbert,
				Lateblue
42	Measurement	9	very late	
	MG	3	early	
		5	medium	
		7	late	
43	Measurement	1	very early	
	MG	3	early	Avonblue,
				Earliblue,
				Weymouth
		5	medium	Herbert,
				Jersey
		7	late	Briteblue,
				Elliott,
				Tifblue
44	Measurement	9	very late	
	MG	3	short	Darrow,
				Northblue
		5	medium	Berkeley,
				Bluecrop
		7	long	Briteblue,
				Tifblue, Woodard
45	Measurement	3	early	
	MG			
		5	medium	
		7	late	

TABLE 1-continued			
(Comparison of characteristics among varieties)			
The present variety		Control Varieties	
Charact. No	RYOKU NH-11	Blueray	Chandler
1	7	6	7
2	7	5	7
3	1	1	2
4	5	5	4
5	6	5	7
	(214 mm)	(173 mm)	(288 mm)
6	4	3	5
	(15.3 mm)	(14.0 mm)	(167 mm)
7	5	5	5
	(69.4 mm)	(68.0 mm)	(67.0 mm)
8	5	5	5
	(35.2 mm)	(31.0 mm)	(36.3 mm)
9	5	5	5
	(157)	(2.20)	(185)
10	3	3	3
11	5	5	5
12	2	2	2
13	5	5	5
14	1	1	1
15	5	7	5
16	6	5	6
	(35.6 mm)	(28.7 mm)	(35.1 mm)
17	2	1-2	2-3
18	2	3	1
19	7	5	7
20	1	3	1
21	9	9	9
22	5	7	7
23	3	7	3
24	7	5	7
	(4.51 g)	(2.89 g)	(5.72 g)
25	3	3	3

TABLE 1-continued			
(Comparison of characteristics among varieties)			
26	5	7	7
	(2.54 mm)	(3.80 mm)	(3.98 mm)
27	2	2	2
28	3	3	3
29	3	1	3
30	7	5	7
	(7.18 mm)	(4.40 mm)	(6.46 mm)
31	7	7	7
	(2.40 mm)	(2.60 mm)	(2.32 mm)
32	5	5	5
33	2	2	2
34	5	6	5
35	2-3	3	3
36	5	7	6
	(Bx 10.8)	(Bx 12.6)	(Bx 13.0)
37	7	5	6
	(pH 2.80)	(pH 3.12)	(pH 2.96)
38	1	1	1
39	3	3	3
40	5	5	5
	Apr. 8 (2016)	Apr. 6 (2016)	Apr. 8 (2016)
41	6	5	6
	Apr. 26 (2016)	Apr. 23 (2016)	Apr. 27 (2016)
42	—	—	—
43	6	5	7
	Middle July to late July	Middle July to late July	Late July to early August
44	5	5	5
45	—	—	—

What is claimed is:
1. A new and distinct variety of *Vaccinium corymbosum* L. plant named ‘RYOKU NH-11’, as described and illustrated herein.

* * * * *

Fig. 1



Fig. 2



Fig. 3

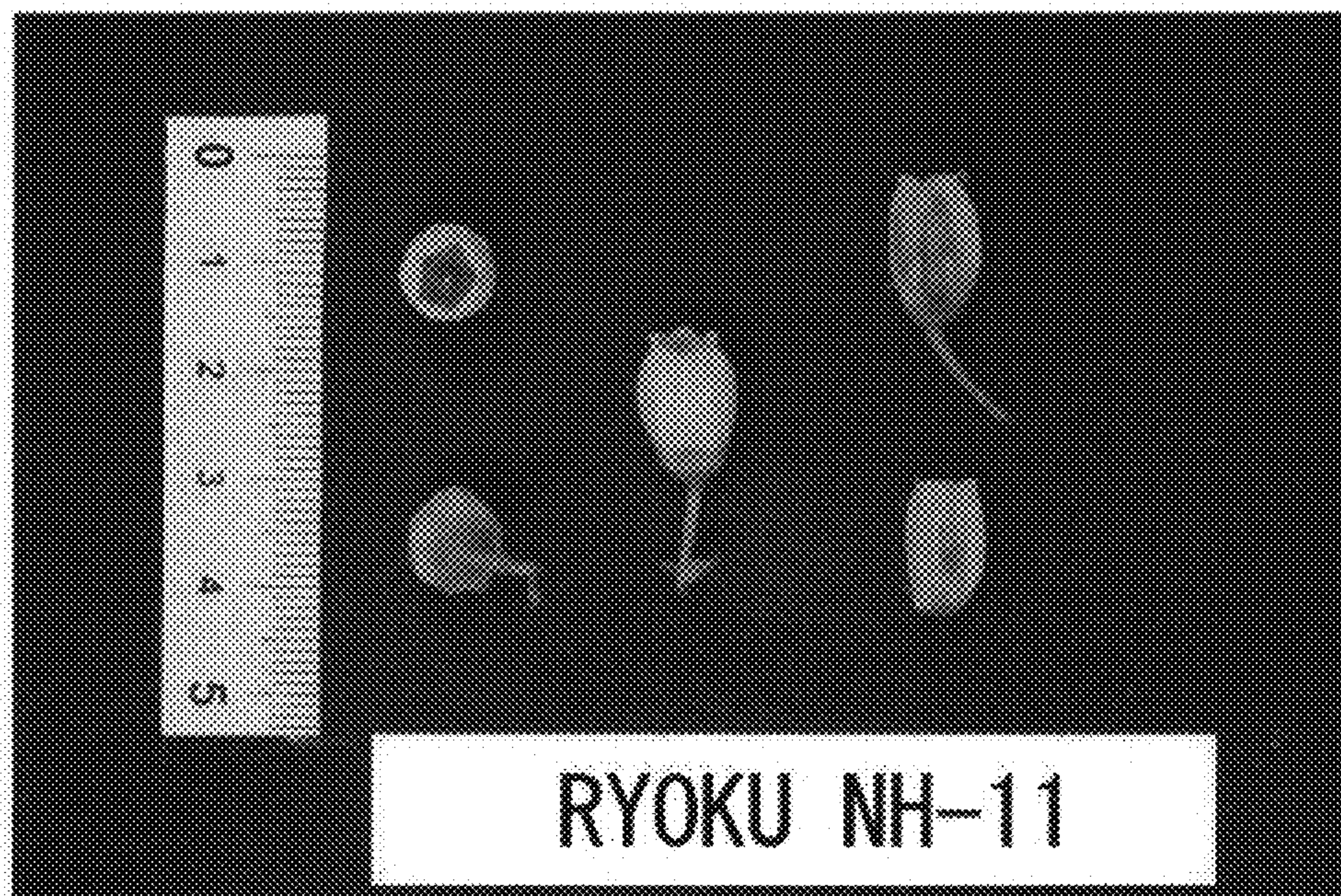


Fig. 4

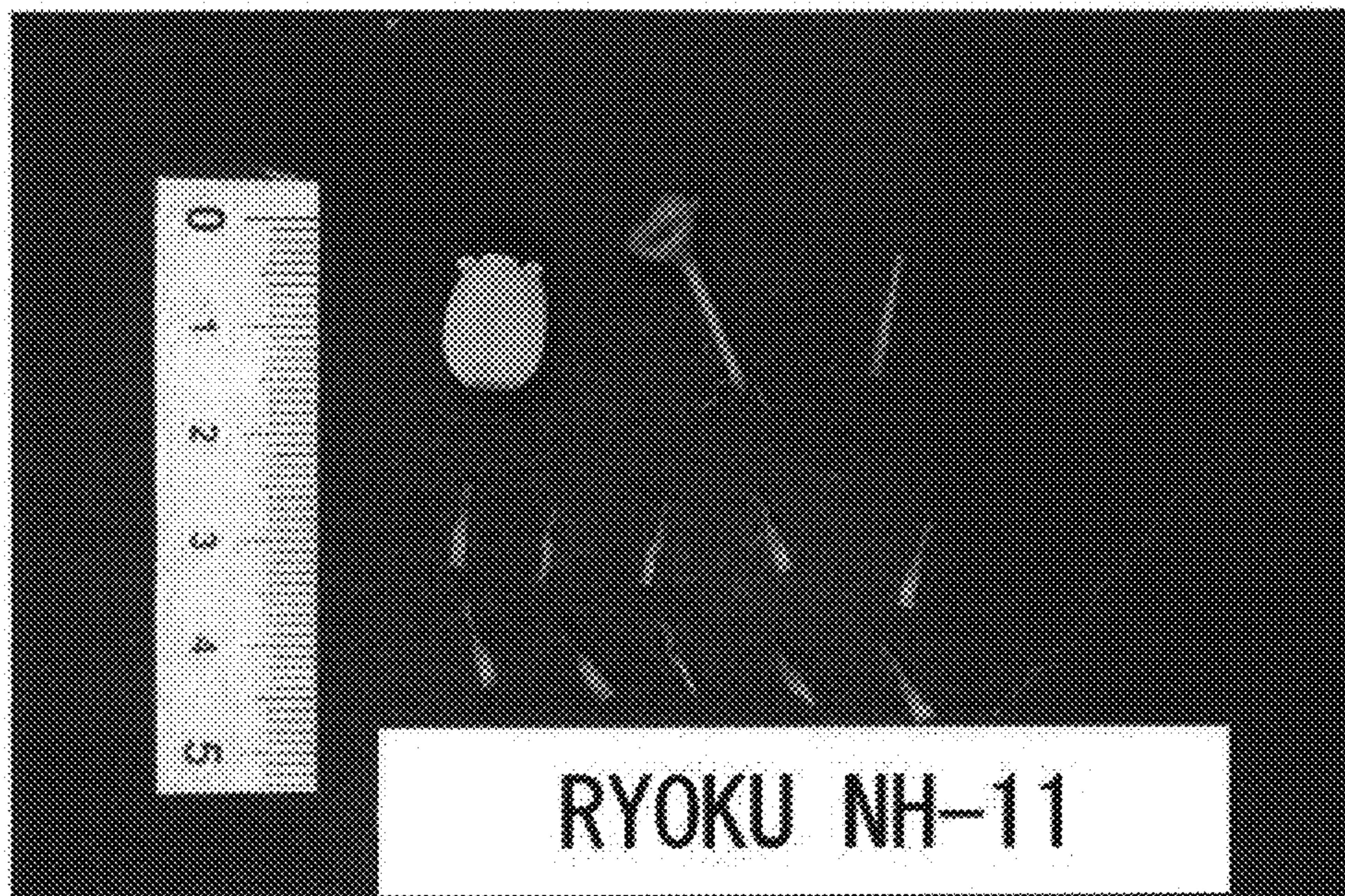


Fig. 5

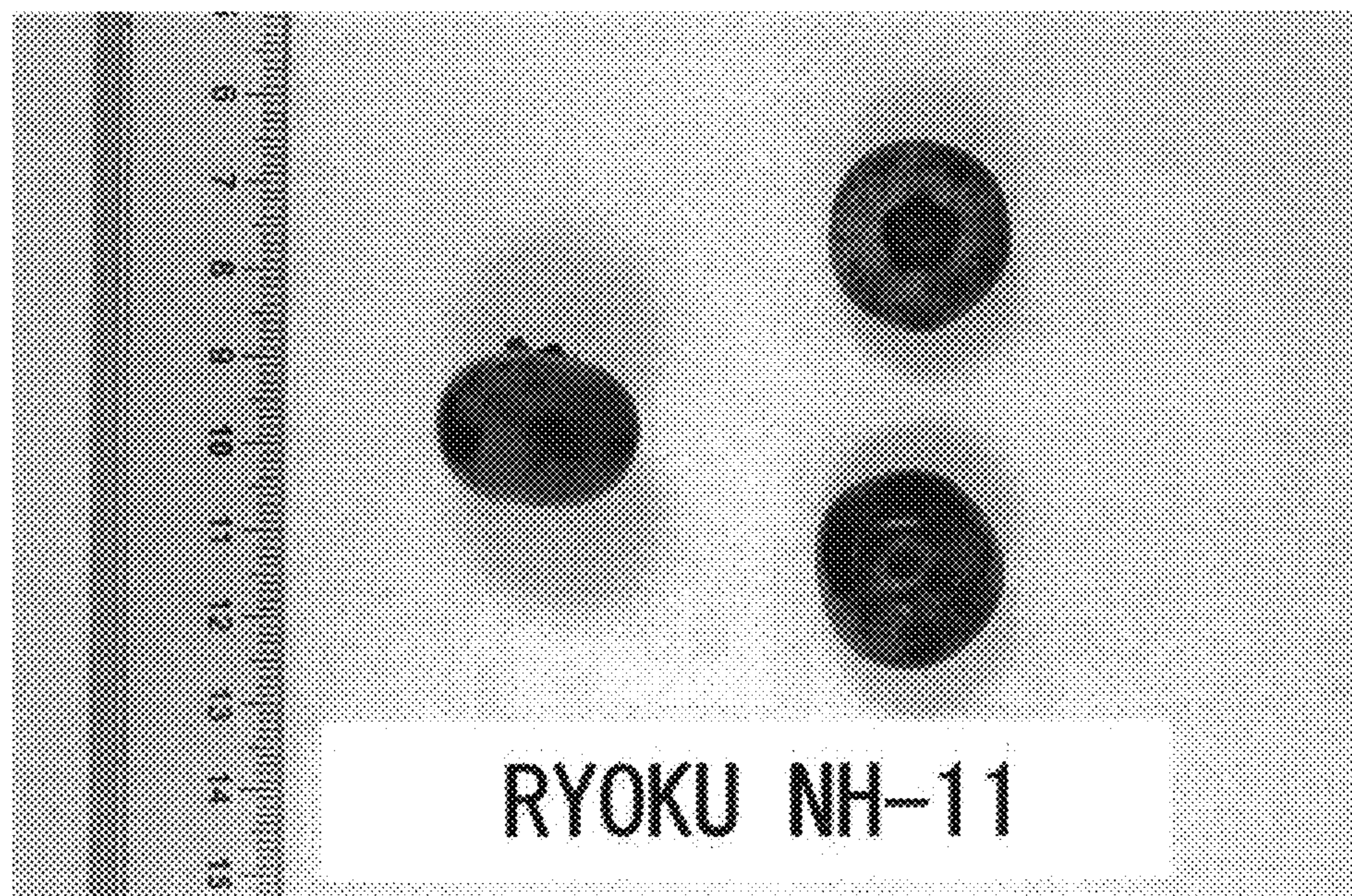


Fig. 6

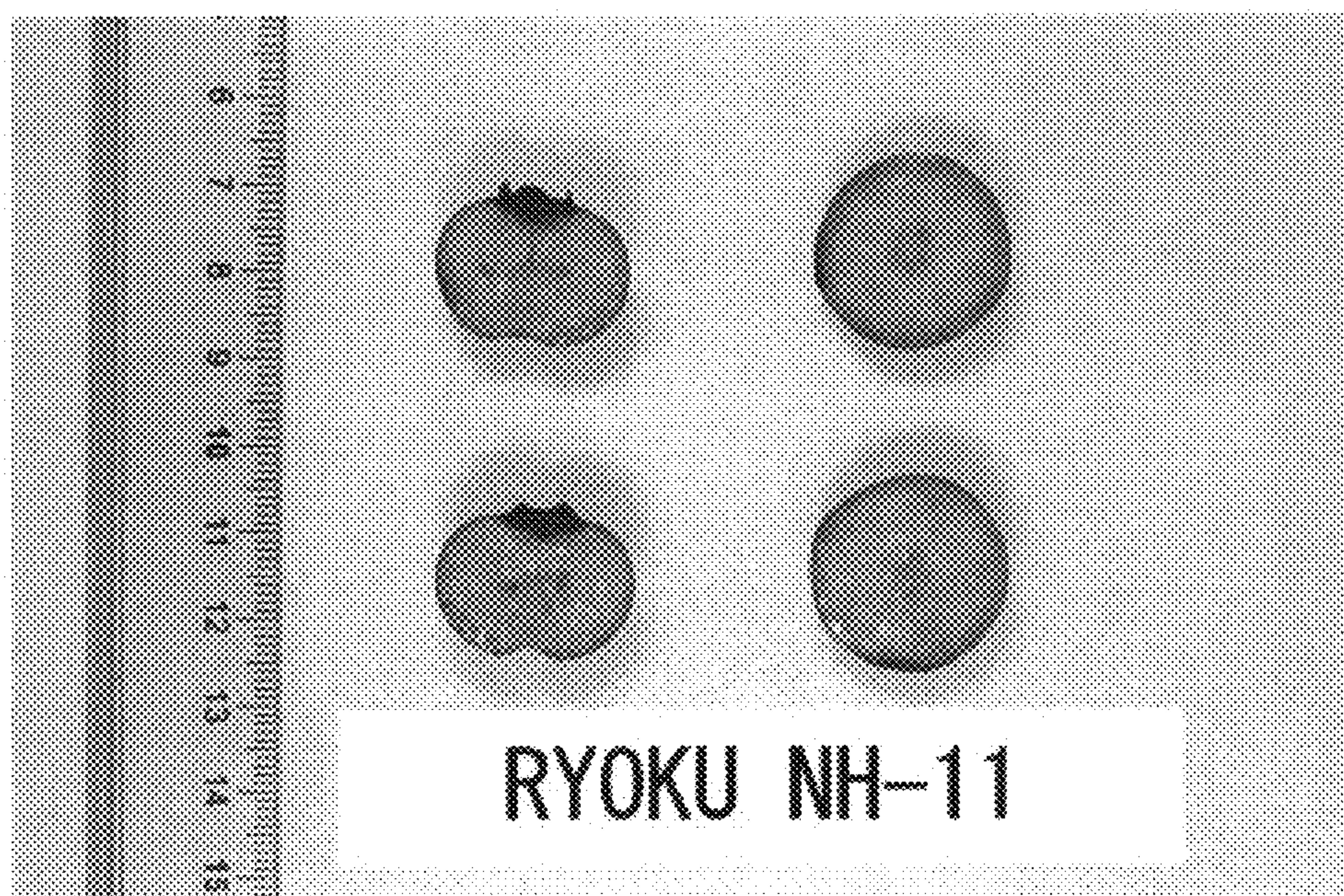
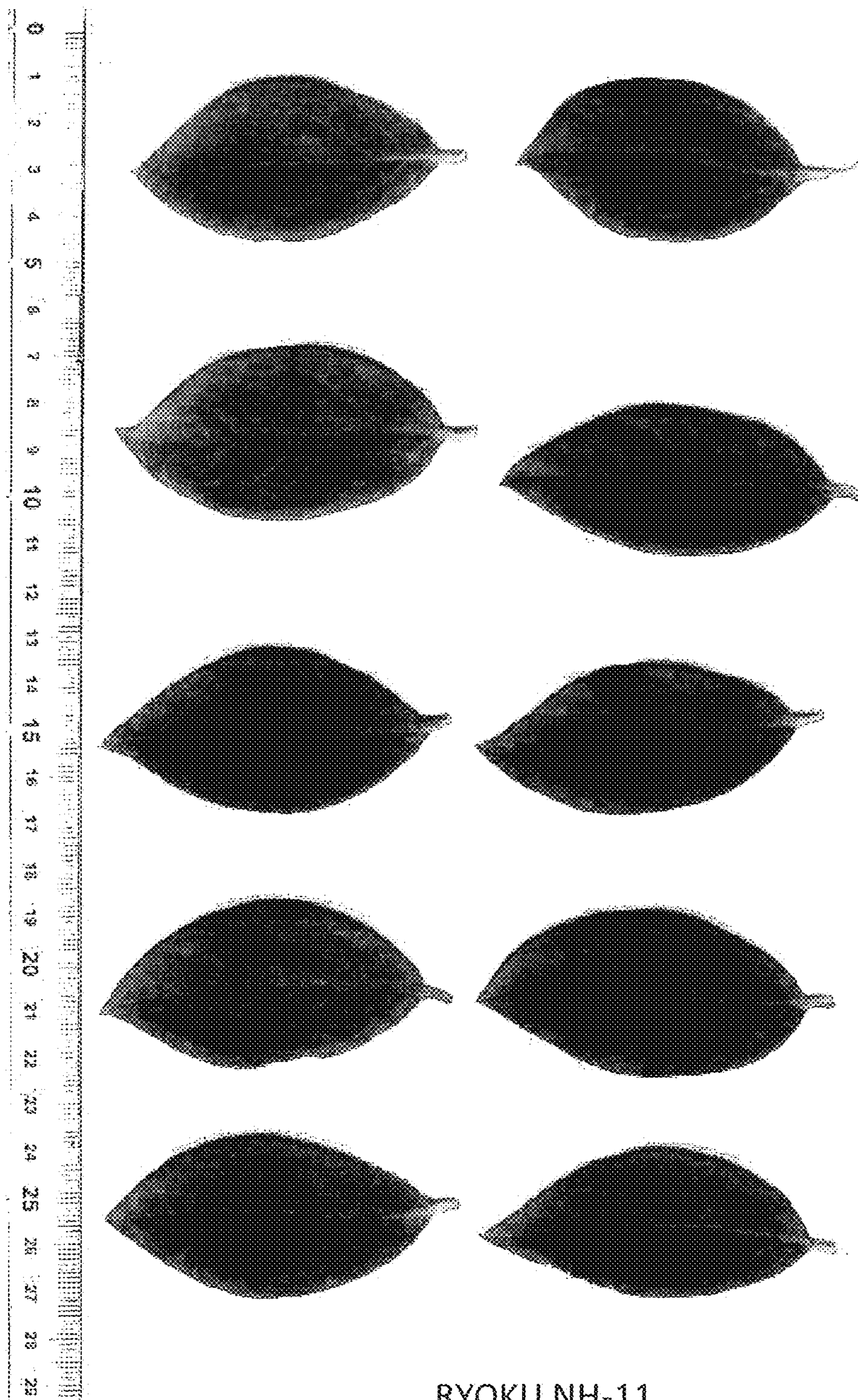
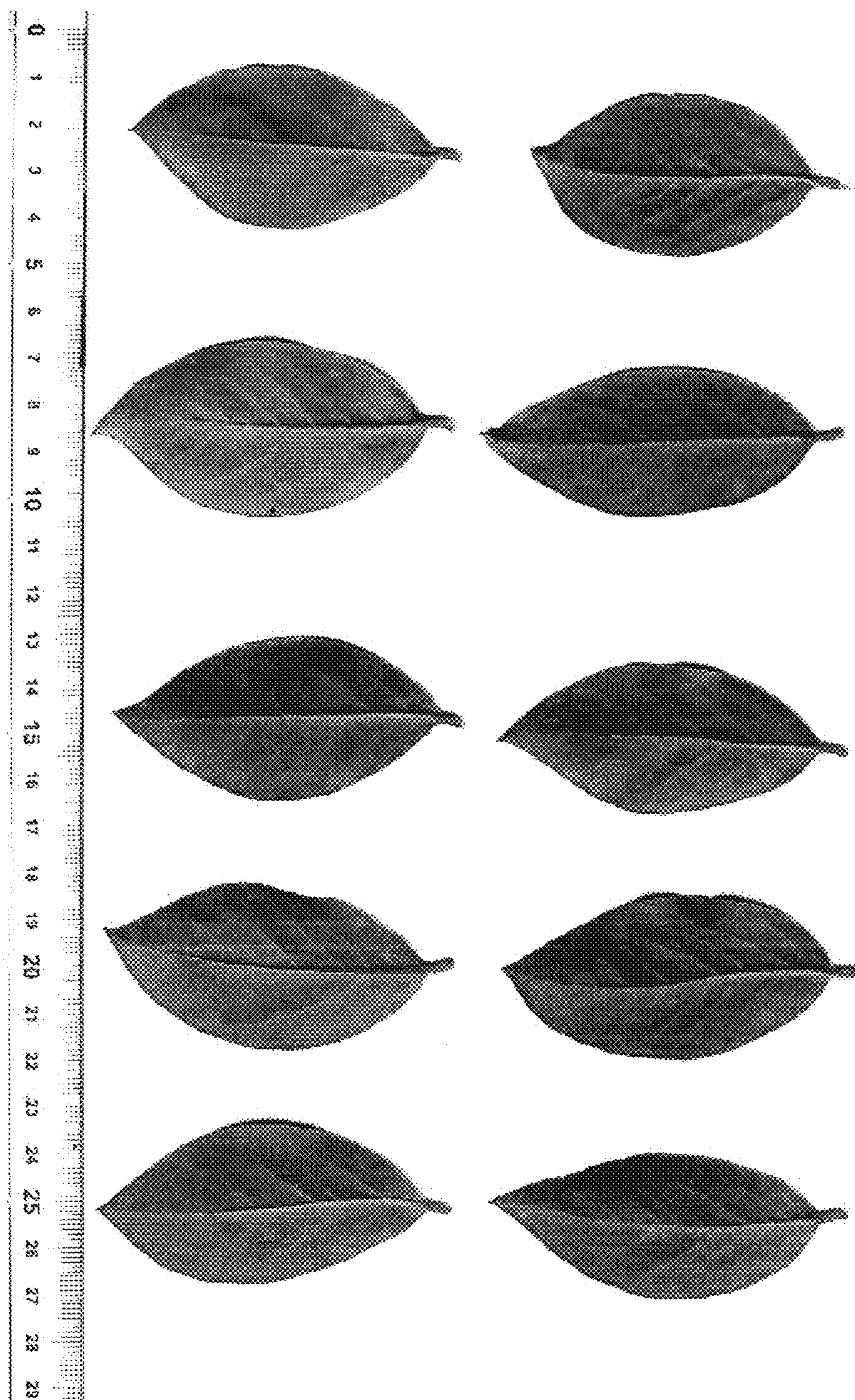


Fig. 7



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Fig. 8



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